DSA03-Natural Language Processing

List of Lab Program

9. Implement a rule-based part-of-speech tagging system using regular expressions using python.

Code:

```
import nltk
from nltk import RegexpTagger, word_tokenize
nltk.download('punkt_tab', quiet=True)
text = "The clever boy plays cricket"
patterns = [
    (r'The|the|A|a|An|an$', 'DT'),
    (r'.*ly$', 'RB'),
    (r'.*ing$', 'VBG'),
    (r'.*ed$', 'VBD'),
    (r'plays|runs|eats$', 'VBZ'),
    (r'boy|cricket$', 'NN'),
    (r'.*', 'NN')
]
tagger = RegexpTagger(patterns)
words = word_tokenize(text)
tags = tagger.tag(words)
print("Word\t\tPOS Tag")
print("-" * 25)
for word, tag in tags:
```

```
print(f"{word:<16}{tag}")
```

Output:

```
Word            POS Tag
-------------------------
The             DT
clever          NN
boy             NN
plays           VBZ
cricket         NN
```